

Decoupling-Oriented Coordination of Paralleled VSGs With Multi-Agent Reinforcement Learning

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Abstract—Paralleled virtual synchronous generators (VSGs) must support system frequency while damping relative motion among units. When a learned controller changes total inertia and its spatial allocation together, an apparent coordination gain can instead come from extra aggregate support or a favorable seed. We separate sustained common-frequency restoration, a 3 s common-inertia pulse, and differential inertia redistribution. The first two layers are fixed; a parameter-shared multi-agent policy is restricted to a bounded, slew-limited, hard-zero-sum two-area residual, benchmarked against a size-matched centralized actor under the same scalar action space. Both architectures use three predefined training seeds and are fixed before an independent 24-disturbance evaluation bank is generated. Across 192 matched 60 s trajectories, the centralized actor reduces synchronization loss by 24.35 % and early inter-area error by 17.04 %; the shared policy reduces them by 16.79 % and 9.54 %. The 95 % intervals account for both seed and disturbance variability and exclude zero for all four effects. However, the shared policy is worse than the centralized actor by 9.98 % and 9.04 %, with all three seeds agreeing. All action and storage constraints are satisfied. Thus, constrained learned differential allocation transfers to the independent evaluation bank, but the tested shared scalar factorization remains inferior to centralized TD3.

Index Terms—virtual synchronous generator, multi-agent reinforcement learning, grid-forming converter, virtual inertia, frequency coordination

I. INTRODUCTION

Virtual synchronous generators (VSGs) reproduce swing-like behavior in converter controls and provide configurable virtual inertia and damping [1], [2]. When several VSGs operate in parallel, however, their average frequency and their relative motion are different physical coordinates. Raising all inertias can improve the common transient, whereas changing their spatial allocation can alter inter-unit and inter-area motion. A controller that treats these effects as one score may improve system frequency without showing improved coordination among the VSGs.

Classical work addresses complementary parts of this problem. Consensus-based secondary control restores frequency in parallel VSG systems [3], [4], while mutual-damping designs target power oscillations and relative modes [5], [6]. These methods make the need for both restoration and damping clear, but they do not by themselves answer a narrower allocation question: after the required common support is fixed, is a disturbance-dependent differential inertia adjustment still useful? This distinction matters because changing total

support is usually an easier route to a lower frequency-error metric than learning where that support should be placed.

Recent learning-based VSG controllers adapt inertia and damping using single-agent soft actor-critic [7], multi-agent deep deterministic policy gradient [8], or other multi-agent policies [9], [10]. Recent studies also coordinate virtual inertia with load-frequency control through MARL [11] and combine learned inertia-damping adaptation with explicit power-decoupling control [12]. These studies support model-free adaptation under changing operating conditions. Nevertheless, jointly learning several inertia-damping actions can mix aggregate support, differential redistribution, and action effort; selecting the best checkpoint can also give an optimistic estimate if training-seed variability and a weak reference are not separated from the claimed physical mechanism.

The submitted digest reported a DDIC reproduction and an inference-time weighted actor ensemble. A subsequent controlled reconstruction did not isolate an ensemble effect beyond selection of a favorable actor. The full paper therefore omits that component and evaluates the differential coordinate against matched causal and centralized baselines.

Evaluating power-system reinforcement learning (RL) therefore requires both structural constraints and explicit treatment of uncertainty. Safety filters and Lyapunov-based methods illustrate how learned controls can be restricted by physical requirements [13], [14]; statistical studies of RL also show that point estimates alone can reverse conclusions in few-run settings [15].

This paper asks two questions: after sustained restoration and aggregate transient support are fixed, does learned differential allocation transfer to unseen disturbances, and how does the tested shared scalar factorization compare with a size-matched centralized actor? The controller is divided by both timescale and coordinate. A slow droop-PI storage layer restores common frequency, a short common-inertia pulse supplies aggregate fast support, and learning can change only one two-area residual. A deterministic projection enforces equal actions within each area and exact normalized zero-sum action across the four VSGs.

The study makes three contributions. First, a common/differential and fast/slow decomposition isolates the value of differential learning. Second, a hard-zero-sum shared policy is benchmarked against a size-matched centralized actor under the same scalar action space. Third, both architectures are fixed

before evaluation on 24 new disturbances with three predefined training seeds and full 60 s trajectories; the reported intervals account for both seed and disturbance variability. Accordingly, this is a controlled test of one shared scalar factorization, not a claim for or against multi-agent reinforcement learning (MARL) as a class.

II. DECOUPLED CONTROL PROBLEM

A. Common and differential coordinates

At the electromechanical timescale, VSG i is represented by

$$M_i \Delta \dot{\omega}_i + D_i \Delta \omega_i = \Delta p_{m,i} - \Delta p_{e,i}, \quad (1)$$

where $\Delta \omega_i = (f_i - f_0)/f_0$, and the two powers are on the 100 MVA system base. In ANDES, $M_i = 2H_i$ is the GENCLS machine start-up time in seconds and D_i is its per-unit damping coefficient; each VSG proxy has $S_n = 200$ MVA. The normalized actions are converted to these coefficients and written directly to GENCLS.M/D every 0.2 s, with interpolation over the internal simulation steps. These are phasor-domain proxy parameters, not hardware ratings. The nonlinear model, rather than (1), generates every reported trajectory.

For $\Delta \mathbf{f} = [\Delta f_1, \dots, \Delta f_4]^\top$, define

$$\Delta f_c = \frac{1}{4} \mathbf{1}^\top \Delta \mathbf{f}, \quad (2)$$

$$\Delta f_{AB} = \frac{1}{2} (\Delta f_1 + \Delta f_2) - \frac{1}{2} (\Delta f_3 + \Delta f_4), \quad (3)$$

$$L_{\text{sync}} = \frac{1}{4} \sum_{i=1}^4 (\Delta f_i - \Delta f_c)^2. \quad (4)$$

Here $\Delta f_i = f_0 \Delta \omega_i$ is reported in hertz. Equations (2)–(4) separate common and differential motion. The common coordinate measures restoration and aggregate excursions; Δf_{AB} and L_{sync} measure differential motion. The control design prevents a learned reduction in (4) simply by increasing every M_i .

B. Fixed common layers

The slow layer requests equal active power from four storage devices:

$$P_{i,k}^* = \Pi_{\mathcal{F}} \left[K_p (f_0 - f_{c,k}) + K_i \sum_{\tau=0}^k (f_0 - f_{c,\tau}) \Delta t \right], \quad (5)$$

In (5), $K_p = 2.0$ system-pu/Hz/device, $K_i = 0.2$ system-pu/(Hz s)/device, and $\Pi_{\mathcal{F}}$ enforces power, ramp, converter-capability, state of charge (SOC), and energy limits with conditional anti-windup. These gains were selected in the preceding development stage and held fixed for the 24-case evaluation; they are not claimed as universal optima. This layer is responsible for sustained restoration.

The fast common layer is a fixed normalized pulse $c_k = 0.25$ for the first 15 samples (3 s) and zero afterwards. With $M_0 = 200$, $\Delta M_{\text{max}} = 600$, and fixed $D_i = 100$, the common-only reference is $M_i = M_0 + \Delta M_{\text{max}} c_k$. The slow layer is evaluated against zero support, and the pulse then against the slow-only reference before learning is introduced;

a development-stage outcome-seeing screen preferred the selected pulse in five of eight cases, after which it was fixed.

Figure 1 summarizes the fixed common layers, differential coordinate, and matched centralized alternative.

III. HARD-ZERO-SUM LEARNING ARCHITECTURES

A. Executed action

One memoryless actor μ_θ is shared by four agents. Each invocation maps a seven-dimensional local observation to $z_{i,k} \in [-1, 1]$. The execution layer aggregates the four votes as

$$q_k^{\text{raw}} = \frac{q_{\text{max}}}{2} [\text{mean}(z_{1,k}, z_{2,k}) - \text{mean}(z_{3,k}, z_{4,k})], \quad (6)$$

$$q_k = \text{clip}(q_k^{\text{raw}}, q_{k-1} - \Delta q_{\text{max}}, q_{k-1} + \Delta q_{\text{max}}), \quad (7)$$

$$\mathbf{r}_k = q_k [1, 1, -1, -1]^\top, \quad (8)$$

A final clip enforces $|q_k| \leq q_{\text{max}}$, where $q_{\text{max}} = \Delta q_{\text{max}} = 0.25$. During the active window,

$$M_{i,k} = M_0 + \Delta M_{\text{max}} (c_k + r_{i,k}), \quad D_{i,k} = 100. \quad (9)$$

After 3 s, $c_k = \mathbf{r}_k = \mathbf{0}$. Because the magnitude and slew constraints act on the scalar q_k , within-area equality and $\mathbf{1}^\top \mathbf{r}_k = 0$ cannot be broken by elementwise clipping. Equations (6)–(8) define the scalar projection; the physical mapping in (9) is checked separately for finite-precision error.

Each agent observes local frequency deviation, common frequency deviation, local-minus-common deviation, local rate of change of frequency (RoCoF), its previous signed residual, local electrical-power change since reset, and a fixed area sign. Each component is scaled in physical units and clipped to $[-5, 5]$. Thus the actors share parameters but receive different local measurements; execution still requires the small coordination layer in (8) to aggregate their votes.

B. Matched training objective

The shared controller uses one 64–64 actor (4737 parameters) at all four agents and one pair of centralized twin-delayed deep deterministic policy gradient (TD3) critics [16]. A matched centralized controller maps the concatenated 28-dimensional observation directly to q_k through a 55–55 actor (4731 parameters). Both controllers use the same critic architecture and hyperparameters, execution map, training data, optimizer settings, replay schedule, exploration process, and team reward. The critics receive the joint observation and executed scalar q_k/q_{max} , following the centralized-critic principle [17]. The reward is

$$R_k = -\frac{L_{\text{sync},k}}{2(0.05)^2} - \frac{\Delta f_{AB,k}^2}{2(0.05)^2} - 0.01 \left(\frac{q_k - q_{k-1}}{0.25} \right)^2. \quad (10)$$

No terminal-frequency, RoCoF, or storage term is included in (10); those quantities are assessed independently.

Critics use 64–64 hidden units; both learning rates are 3×10^{-4} . The discount, target-update factor, replay capacity, batch size, and warm-up are 0.99, 0.005, 10^5 , 256, and 512 transitions. Policy noise, clipping noise, exploration noise, and the delayed update period are 0.1, 0.2, 0.1, and 2. Seeds 17, 53,

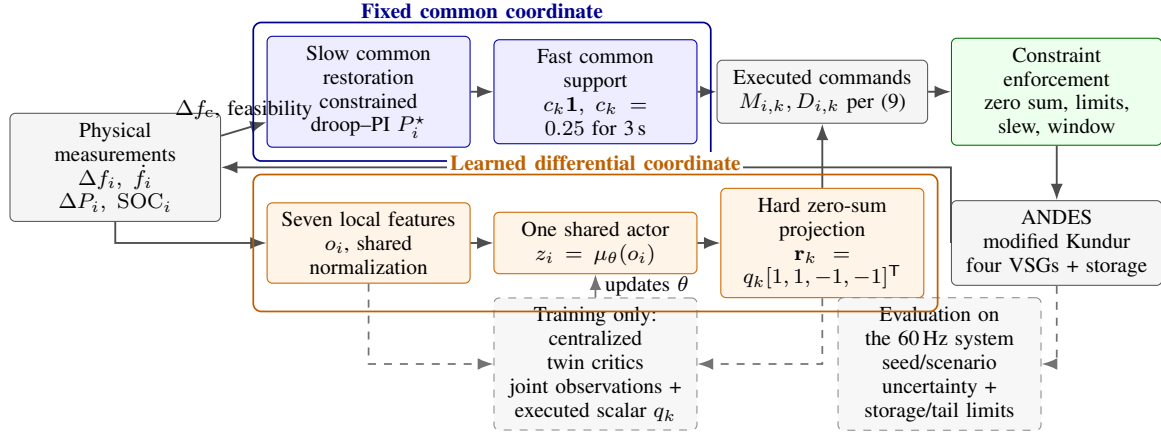


Fig. 1. The control architecture separates aggregate support from differential coordination. Fixed layers regulate the common coordinate; the shared policy can alter only one bounded two-area inertia coordinate whose normalized fleet sum is zero by construction. The size-matched centralized baseline replaces the shared actor and projection with one joint-observation actor producing the same scalar q_k . Dashed paths are used only during centralized training and evidence collection.

and 89 are each trained for 300 episodes of 15 steps. Only the six final checkpoints are evaluated; there is no best-checkpoint selection or early stopping.

C. Genuine three-edge comparison

A separate sealed comparison on the same plant evaluates a genuinely distributed formulation: one shared actor is invoked separately on each of the three path edges using only that edge's two endpoint observations, and its three scalar outputs are the three executed edge actions, with the joint critic training-only. A matched joint-observation single actor maps the concatenated observation to the identical three action coordinates. Both learned arms share the selected classical edge prior, projection, limits, reward, training budget, five predefined seeds, bank, and thresholds, and the selected deterministic classical edge controller is retained as the strong comparator. This identifies only the executed endpoint-only and joint-observation information patterns, not MARL as a class.

IV. EXPERIMENTAL SETUP AND EVALUATION

A. Plant and disturbance bank

All experiments use ANDES 2.0.0 [18] and a modified two-area Kundur system [19] with four VSG proxies. Each PV+GENCLS proxy is paired with an independent grid-following ESD1 storage device at buses 12, 16, 14, and 15; this arrangement is not a unified physical grid-forming battery energy storage system. Each storage device is rated 36 MVA/28 MWh, has a 0.36-system-pu power limit, a one-second external ramp, and SOC limits of 0.2–0.8 from an initial 0.5. The rating is a 50-fold system-study scaling of the 720 kVA/560 kWh experimental unit reported in [20]; the explicit SOC, converter, and ramp limits follow storage-modeling practice summarized in [21].

The 24-case development bank covers four disturbance locations, both signs, and moderate (0.65–0.85 pu), strong (0.95–1.15 pu), and edge (1.35–1.50 pu) magnitudes, and is used for

classical checks and training only. After all six controllers are fixed, an independent generator (seed 2026072704) creates the evaluation bank with the same balanced location/sign/severity design but new magnitudes. A zero-support screen that does not use controller outcomes accepts all 24 cases; no case is redrawn or excluded. The control interval is 0.2 s. Training uses the 3 s active window, whereas every evaluation trajectory runs for 60 s and reports frequency in the physical units of the 60 Hz system.

B. Physical checks and evaluation rule

Before testing learned controllers, we verify each physical prerequisite. The slow-layer check compares droop-PI storage with zero requested storage power. The fast-layer check adds the common-inertia pulse above the slow layer. A four-arm comparison of zero support, the slow layer, the fast layer, and both layers tests whether their benefits are additive or synergistic. Finally, an outcome-seeing oracle selects the best of six fixed differential pulses or the reference for each case. Because this selection uses future information, the oracle provides only an upper bound on attainable differential benefit.

The evaluation-bank test compares eight arms: the fixed common layers with $q = 0$, a bounded causal area-RoCoF baseline, three centralized TD3 seeds, and three shared-MARL seeds, giving 192 scenario-matched trajectories. The causal controller uses only the current joint observation:

$$q_k^c = q_{\max} \tanh \left[-k_f \frac{\Delta f_{AB,k}}{0.05} - k_r \frac{\Delta \dot{f}_{AB,k}}{0.5} \right]. \quad (11)$$

The denominators are in hertz and hertz per second. Nine gain pairs were fixed before the development comparison; among candidates meeting all checks, the lowest worst-location score selected $(k_f, k_r) = (0, 0.25)$. This law was fixed before the independent bank and passed through the same magnitude, slew, active-window, and physical-action mapping as the learned controllers.

The two primary endpoints are the full-trajectory mean of (4) and $\sum_{k=1}^{15} |\Delta f_{AB,k}| \Delta t$. Secondary checks are maximum sampled RoCoF, worst-bus peak, full-horizon common-frequency integral absolute error (IAE), and mean common error in the final 10 s. A controller is effective only if both primary effects improve by at least 2% with both 95% interval upper bounds below zero; fast metrics may not worsen by more than 5%, and slow metrics by more than 2%. Completion and limits on action magnitude, slew, active-window compliance, storage power and energy, SOC, saturation, constraints, and empirical 90% conditional value at risk (CVaR90) must also pass.

The causal baseline uses 10 000 paired-bootstrap resamples of matched disturbances. For learned controllers, each of 10 000 resamples first draws from the three training seeds and then from the shared disturbance cases, so the intervals include both sources of variability. Per-seed effect directions are also reported to show agreement or disagreement across seeds.

V. RESULTS AND DISCUSSION

Figure 2 summarizes the classical mechanism checks and the evaluation-bank architecture comparisons.

A. Classical authority and attainability

The slow layer materially restored the common coordinate. Relative to zero support, it reduced full-horizon mean-frequency IAE by 58.63 % (95% interval -58.67% to -58.58%) and final-window common error by 77.29 % (-77.36% to -77.22%); all 24 pairs improved on both endpoints. Above this established slow layer, the common-inertia pulse reduced maximum RoCoF by 28.37 %, worst-bus peak by 11.08 %, synchronization loss by 4.29 %, and early inter-area IAE by 9.91 %, with all paired intervals below zero.

The four-arm comparison showed additive rather than synergistic behavior: none of six interaction terms reached the required -2% improvement with an interval below zero (estimates range from -0.15% to $+1.49\%$ of the zero-support mean). Hence the two classical layers are worth retaining, but their joint value is not evidence of a special coordination synergy.

A separate fresh-bank handoff gate challenged the fixed 3 s timing. Neither a 5 s pulse nor either of two causal state-aware smooth handoffs cleared the preregistered joint post-handoff gate relative to the fixed 3 s benchmark. The common-state rule traded lower post-window frequency error for a higher secondary peak, while the added differential and slow-layer-gap conditions had no material incremental value. Thus the fixed timing is retained as a bounded benchmark, not as a solution of fast-slow coupling.

The outcome-seeing oracle then tested whether any differential action could still help. It chose a non-reference differential pulse in 18 of 24 cases and reduced synchronization loss by 25.64 % and early inter-area IAE by 19.66 %, with paired intervals wholly below zero. This establishes an attainable differential margin on the development bank; it does not establish that a causal policy can identify the action.

TABLE I
EVALUATION-BANK EFFECTS ON THE TWO PRIMARY ENDPOINTS

Comparison	Endpoint	Effect (%)	95% CI (%)
Central/ $q = 0$	Sync loss	-24.35	[-31.50, -17.48]
Central/ $q = 0$	Area IAE	-17.04	[-24.60, -8.96]
Shared/ $q = 0$	Sync loss	-16.79	[-24.73, -5.81]
Shared/ $q = 0$	Area IAE	-9.54	[-17.19, -0.60]
Shared/central	Sync loss	+9.98	[+1.04, +26.13]
Shared/central	Area IAE	+9.04	[+3.09, +19.54]

B. Evaluation-bank comparison of learned controllers

The bounded causal area-RoCoF baseline in (11) met the criterion on neither endpoint. Its synchronization-loss effect was $+0.03\%$ (95% interval $[-0.26, +0.43]\%$); its area-IAE effect was -0.09% ($[-0.19, +0.03]\%$). Table I shows a different result for learned policies. Centralized TD3 reduced synchronization loss by 24.35 % and area IAE by 17.04 %. Shared MARL also met the criterion on both endpoints, with 16.79 % and 9.54 % reductions. These intervals include both seed and scenario variability and exclude zero.

The direct architecture comparison showed that shared MARL was worse than the centralized actor by 9.98 % in synchronization loss and 9.04 % in area IAE. The intervals lie entirely above zero, and each of the three paired seeds favors centralized TD3 on both endpoints, so the learned differential action is directionally consistent across seeds. Thus, the tested shared scalar factorization improved on $q = 0$ but was inferior to centralized TD3; broader multi-agent designs were not tested.

All 192 trajectories completed 300 samples. Maximum commanded storage power was 0.333022 pu, SOC remained in $[0.485184, 0.511006]$, and no saturation reason or constraint violation occurred. Action magnitude, slew, active-window, relative no-harm, and storage limits were all satisfied; the hard-zero-sum map prevents either learned controller from obtaining differential improvement by increasing fleet-average inertia.

Figure 3 makes the coordinate constraint visible. Each learned scalar q_k moves the two area inertias in opposite directions while their fleet mean remains equal to the reference allocation. Common frequency and aggregate storage power remain close to $q = 0$, so the illustrated differential response is not produced by additional common support. At 3 s, the common pulse and differential residual are switched off by design; the jump is not closed-loop settling. Inference comes from all 24 scenarios and all three seeds in Fig. 2, not from this trace.

C. Three-edge architecture comparison

A separately sealed fresh bank challenges the three-edge formulation in Section III against the selected classical edge controller. The complete 288-cell matrix (264 new controller trajectories plus 24 zero-residual records) passed every integrity, storage, and action guard. The classical controller first established a strong incremental baseline: relative to

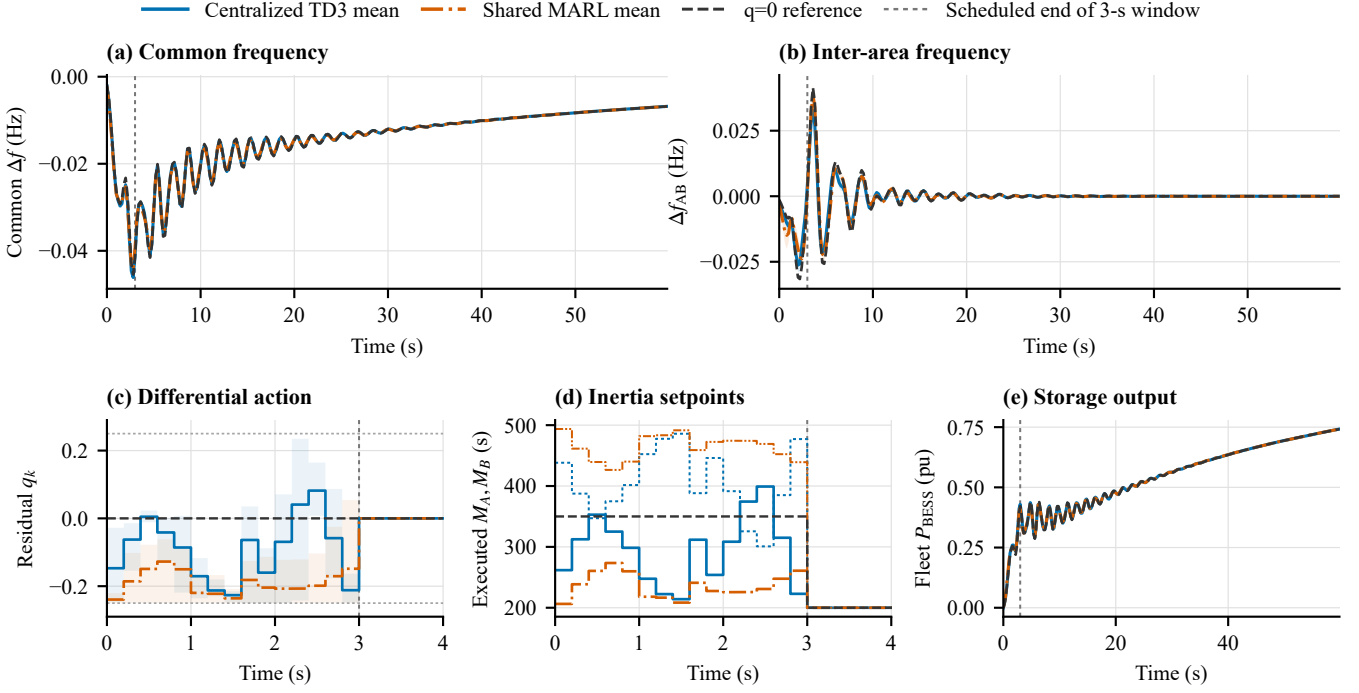


Fig. 3. The fixed illustrative case shows bounded, piecewise-constant differential commands and their scheduled deactivation after 3 s, while common-frequency and fleet-storage responses remain close to the $q = 0$ reference. The case is the first moderate-severity scenario in the fixed evaluation-bank order, selected without using controller outcomes. Lines are three-seed means, shading is the seed range where shown, and panels (c)–(d) use step plots because commands update every 0.2 s; the vertical line marks scheduled deactivation, dotted lines in (c) mark $|q_k| \leq 0.25$, and (d) shows the executed inertia setpoints.

VI. LIMITATIONS AND REPRODUCIBILITY

The independent evaluation bank and three seeds address disturbance and seed robustness within one modified Kundur topology; they do not establish topology generalization, and three seeds do not characterize the wider training-seed population. Because the shared execution layer aggregates four local votes, the result also does not support fully decentralized deployment. Zero-sum projection guarantees fleet-budget neutrality at the input, not modal or dynamic decoupling of the nonlinear plant. The phasor-domain hybrid actuator does not establish electromagnetic-transient, unified-GFM-BESS, motor-dominated-load, stability, or hardware transfer; the missing coupled converter–energy and voltage–frequency dynamics require separate validation. The handoff gate covers only one fresh bank and two deterministic state-aware rules; it neither establishes that 3 s is optimal nor that event-triggered handoff is generally ineffective. The three-edge comparison covers one plant, one fixed four-node path, one sealed 24-scenario bank, and five predefined seeds; its endpoint-only execution and the joint-observation actor differ in information pattern, and its registered trace guards establish empirical feasibility on that bank rather than a stability or safety certificate. It does not show that distributed learning, multi-agent methods, or learned residual control are generally ineffective.

The initial numerical check rejected five of 192 rows because single-precision slew values exceeded 0.25 by at most 7.450581×10^{-9} , less than one unit in the last place,

2.980232×10^{-8} , at that limit. A representation-aware correction changed only those five Boolean checks; all 192 stored trajectory hashes, checkpoints, efficacy statistics, and thresholds were retained. The corrected check and complete experiment provenance are generated directly from the retained artifacts.

VII. CONCLUSION

This paper separates common restoration, common transient support, and differential inertia redistribution in a four-VSG MARL study. The slow droop–PI storage layer restored common frequency, the common-inertia pulse improved the fast transient, and an outcome-seeing oracle confirmed an attainable differential margin. On the independent evaluation bank, both centralized TD3 and shared MARL improved synchronization and early inter-area error across three predefined seeds while satisfying every evaluated physical limit. However, the size-matched centralized actor was consistently stronger. The evidence therefore supports constrained learned differential allocation. In this scalar formulation, parameter-sharing MARL provides meaningful but inferior performance relative to centralized TD3. This does not establish that MARL lacks value generally or that the nonlinear plant is modally decoupled. The tested deterministic handoffs also failed their joint gate, so the fixed 3 s pulse remains a bounded benchmark rather than a general solution of fast–slow coupling. In a separate three-edge formulation, neither the neighbour-only

shared-edge actor nor the matched joint-observation actor improved on the selected classical edge controller, so the classical controller is retained and the negative result is used only as a bounded limitation.

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